Unit 2.2 Printables: Inherited Instructions

Standard: HS-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors. Targets are labeled LS3-2.1 to LS3-2.5 (priority) and LS3-3.1, LS3-3.2 (supporting).

Everything the class needs for this unit is in this one file, organized by lesson. Print on white paper in black ink only. "Copies" says how many to print: per student, per pair, per group or one for the class. Each item is a dashed box; cut cards along the dashed lines. Sources are cited on any item that uses real data; items marked illustrative use simplified numbers made up for class.

Lesson	Printable	Use	Copies
1	Puppy litter data	Glue under "Compare the traits" in the notebook	One per student
2	Allele cards: Parent 1 and Parent 2	Cut apart; each group sorts the cards into 6 piles	One per group of 3 to 4
2	Card table: 3 offspring	Glue under "Make possible offspring" in the notebook	One per student
4	Genotype to phenotype: pea plants	Glue under "Genotype and phenotype" in the notebook	One per student
4	Punnett squares: steps and worked example	Glue under "Punnett squares" in the notebook	One per student
5	Monster Genetics: the coin rule and Table 2	Glue under "How the lab works"; keep for Lesson 6	One per student
5	Monster Genetics sheet: Table 1, the female monster	Lab sheet for the coin-flip build	One per pair
6	Punnett square: skin color and eyes	Glue under "Punnett squares for the baby"	One per student
6	Prediction vs. class count	Glue under "Class data"	One per student
7	Four kinds of mutations	Fill in during the notes; glue in or keep with the notebook	One per student
7	UV light and mutations	Glue under "UV and mutations"	One per student
7	Who passes it on? Five cases	Glue under "Who passes it on?"	One per student
8	Case data: three cases	Glue under "Case 1" (and refer back for Cases 2 and 3)	One per student
9	The new coat pattern: two kennels	Glue under "The new coat pattern"	One per student
10	My target tracker	Glue under "Check my CFAs"	One per student
10	Hydrangea soil data	Glue under "Enrichment: environment and traits"	One per student (enrichment)

Lessons with no printable (3, 11) use the notebook and the slides only.

Puppy litter data

Use: Glue under "Compare the traits" in the notebook · Copies: One per student

Two parents had four puppies. Look at the traits. Circle every trait a puppy shares with Mom. Underline every trait a puppy shares with Dad.

	Fur color	Ear shape	Pattern
Mom	Black	Floppy	Spotted
Dad	Brown	Pointed	Solid
Puppy A	Black	Floppy	Solid
Puppy B	Brown	Floppy	Spotted
Puppy C	Black	Pointed	Spotted
Puppy D	Brown	Floppy	Solid

Finish the sentences

Every puppy has some traits like Mom and some like _____.

No two puppies have the same _____ of traits.

Word bank: combination · Dad

Illustrative data: simplified numbers made up for class, not from a study.

Allele cards: Parent 1 and Parent 2

Use: Cut apart; each group sorts the cards into 6 piles · Copies: One per group of 3 to 4

Each parent has two versions of each of 3 genes. Cut out all 12 cards. Sort them into 6 piles: Parent 1 fur color, Parent 1 ear shape, Parent 1 pattern, and the same 3 for Parent 2. Shuffle each pile. Take one card from every pile to make one offspring. Record it, put the cards back, and repeat.

Parent 1

P1 · Fur color: Black	**P1 · Fur color: Brown**	**P1 · Ear shape: Floppy**
P1 · Ear shape: Pointed	**P1 · Pattern: Solid**	**P1 · Pattern: Spotted**

Parent 2

P2 · Fur color: Black	**P2 · Fur color: Brown**	**P2 · Ear shape: Floppy**
P2 · Ear shape: Pointed	**P2 · Pattern: Solid**	**P2 · Pattern: Spotted**

The cards model sex cells: each parent passes on one version of each gene. Illustrative data: simplified numbers made up for class, not from a study.

Card table: 3 offspring

Use: Glue under "Make possible offspring" in the notebook · Copies: One per student

Offspring 1 is done for you. Make Offspring 2 and 3 with the cards. Write the card you drew from each pile.

Offspring	Fur: from P1	Fur: from P2	Ears: from P1	Ears: from P2	Pattern: from P1	Pattern: from P2
1	Black	Brown	Floppy	Pointed	Solid	Solid
2						
3						

Which offspring look alike? Explain in one sentence.

Genotype to phenotype: pea plants

Use: Glue under "Genotype and phenotype" in the notebook · Copies: One per student

In pea plants, T (tall) is dominant to t (short). TT is done for you. Fill in the table.

Genotype	Phenotype (what it looks like)	Homozygous or heterozygous?
TT	Tall	Homozygous
Tt		
tt		

Tt is _____ because T is dominant. tt is _____.

Word bank: short · tall

Practice: write the genotype

Description	Genotype
Two copies of the dominant allele	
One dominant and one recessive allele	
Short plant	

Pea plant height is the classic Mendel trait (Gregor Mendel, 1866).

Punnett squares: steps and worked example

Use: Glue under "Punnett squares" in the notebook · Copies: One per student

Steps

1. Write one parent's two alleles across the top. 2. Write the other parent's two alleles down the side. 3. Fill each box with the allele from its column and the allele from its row.

Worked example: Tt × Tt

Tt × Tt

	T	**t**
T	TT	Tt
t	Tt	tt

Results: TT, Tt, Tt, tt. That is 3 tall to 1 short.

Your turn

Tt × tt

	T	**t**
t		
t		

TT × Tt

	T	**T**
T		
t		

Each box in a Punnett square shows one possible _____ of alleles. Word bank: combination

Monster Genetics: the coin rule and Table 2

Use: Glue under "How the lab works"; keep for Lesson 6 · Copies: One per student

The coin rule

Flip a coin twice for each trait. Heads = Allele 1 (the capital letter). Tails = Allele 2 (the lowercase letter). Two flips make the genotype. If there is at least one capital letter, the dominant trait shows.

Worked example, Eyes: heads, tails gives Ee. E is dominant, so the monster has two small eyes.

Table 2: the male monster (genotypes are given)

Trait	Dominant (capital)	Recessive (lowercase)	Male genotype	Male phenotype
Eyes	E = two small eyes	e = one large eye	Ee	
Skin color	G = green	g = blue	Gg	
Teeth	T = pointed	t = flat	tt	
Feet	F = webbed	f = round	Ff	
Horn color	H = red	h = yellow	hh	
Claws	K = long	k = short	Kk	
Tail	C = curly	c = straight	cc	
Ears	A = round	a = pointed	AA	
Hair	P = hairy	p = bald	Pp	
Wings	W = wings	w = no wings	ww	
Spots	S = spotted	s = plain	Ss	
Body size	B = large	b = small	bb	

Monster traits and rules are made up for class.

Monster Genetics sheet: Table 1, the female monster

Use: Lab sheet for the coin-flip build · Copies: One per pair

One partner flips and the other records. Switch after 6 traits. Flip twice per trait, write the genotype, then write the phenotype using the dominant rule from your Table 2 key.

Trait	Flip 1	Flip 2	Genotype	Phenotype
Eyes				
Skin color				
Teeth				
Feet				
Horn color				
Claws				
Tail				
Ears				
Hair				
Wings				
Spots				
Body size				

Compare: our monster is different from our partner pair's because the flips gave different _____ of alleles. Word bank: combinations

Predict: when she mates with the male, what is the chance their baby has one large eye? My guess: _____

Punnett square: skin color and eyes

Use: Glue under "Punnett squares for the baby" · Copies: One per student

Worked example together: Skin color, Gg × Gg

Gg × Gg

	G	**g**
G	GG	Gg
g	Gg	gg

Each box is 1 of 4, or 25%. Gg × Gg gives GG, Gg, Gg, gg. Three of 4 boxes are green and 1 is blue.

Now use your own monsters' genotypes (Table 1 and Table 2). Eyes:

Eyes: ____ × ____

The letters on the top and side of a square are the _____ in each parent's sex cells. Word bank: alleles

Percent chance table (6 traits)

Trait	Female × Male	Boxes with the dominant trait	Percent chance (dominant)	Percent chance (recessive)
Eyes				
Skin color				
Teeth				
Feet				
Horn color				
Claws				

UNIT 2.2 · LESSON 6 · CLASS DATA

Prediction vs. class count

Use: Glue under "Class data" · Copies: One per student

Everyone makes one baby: flip a coin for Mom (heads = G, tails = g) and a coin for Dad. Add your baby to the class tally.

Skin color	Punnett prediction	Sample class (30 babies)	Percent in the sample	Our class count
Green (GG or Gg)	75%	22	73%	
Blue (gg)	25%	8	27%	

The Punnett square predicted _____% green. The sample class got about _____% green. The numbers are close but not exact because every flip is _____. Word bank: 73 · random · 75

Compare your own class tally with the prediction. Percent green in our class: _____%

Illustrative data: simplified numbers made up for class, not from a study. The sample class is a made-up result of 30 flips.

Four kinds of mutations

Use: Fill in during the notes; glue in or keep with the notebook · Copies: One per student

DNA is read in groups of three bases called codons. Each codon calls for one amino acid or a stop. Original message (mRNA): AUG GAA CAC UGG. Amino acids: Met - Glu - His - Trp.

Kind	What changes	Example (mRNA)	Effect on the protein
Silent	One base changes, but the new codon calls for the same ____(1)____	GAA -> GAG (Glu -> Glu)	No ____(2)____ on the protein
Missense	One base changes and the codon calls for a ____(3)____ amino acid	GAA -> GUA (Glu -> Val)	One amino acid is different; the protein may work differently
Nonsense	A codon becomes a ____(4)____ codon	UGG -> UGA (Trp -> stop)	Protein-building ____(5)____ early, so the protein is too short
Frameshift	A base is added or deleted, so every codon after it is ____(6)____	AUG GAA CAC UGG -> AUG AAC ACU GG...	Most amino acids after the change are different

Word bank: amino acid · effect · different · stop · terminates · shifted

Codon assignments are from the standard genetic code (e.g., NCBI, "The Genetic Codes"). The example message is made up for class.

UV light and mutations

Use: Glue under "UV and mutations" · Copies: One per student

Skin cells were exposed to different amounts of UV light. Scientists counted new mutations per 1,000 skin cells. Worked example: with 0 hours of UV there are 2 mutations per 1,000 cells. Those come from copying errors.

UV exposure (hours)	Mutations per 1,000 skin cells
0	2
1	6
2	11
4	23

Graph it: put UV hours on the x-axis and mutations on the y-axis.

Graph area

With more UV light, there are _____ mutations. The 2 mutations with no UV came from _____ errors. Word bank: copying · more

Illustrative data: simplified numbers made up for class, not from a study. Pattern modeled on the known effect of UV light on DNA damage in skin cells.

Who passes it on? Five cases

Use: Glue under "Who passes it on?" · Copies: One per student

Could the next generation inherit each change? Case A is done for you.

Case	Where did the change happen?	Could the next generation inherit it? (Yes or No)	Why
A	A skin cell gets a UV mutation and becomes a mole.	No	It is a body cell, not an egg or sperm.
B	A copying error happens in a cell that makes sperm.		
C	UV light mutates a skin cell of a dog.		
D	A mutation occurs in an egg cell before fertilization.		
E	A chemical mutates a liver cell.		

Why: a change in a _____ cell is not passed on. A change in a cell that makes _____ can be. Word bank: eggs or sperm · body

Explain it: How can UV light change a gene, and when is the change inherited?

_____

_____

_____

Case data: three cases

Use: Glue under "Case 1" (and refer back for Cases 2 and 3) · Copies: One per student

Case 1: the puppy litter

Both parents are Bb. B is black, b is brown. The litter has 3 black puppies and 1 brown puppy. The brown puppy is bb.

Case 2: a new base (coat-color gene)

	Allele 1	Allele 2
Parent 1	CCA	CCG
Parent 2	CCA	CCC
Puppy	CCA	CCT

Each puppy gets one allele from each parent. The letters show a short stretch of DNA in the coat-color gene.

Case 3: X-rays and mice

X-ray dose	New mutations per 1,000 sperm cells
None	3
Low	9
High	25

Scientists exposed male mice to X-rays and counted new mutations in sperm cells.

Illustrative data: simplified numbers made up for class, not from a study. Background: radiation raises mutation rates in mouse sperm cells (e.g., W. L. Russell's mouse studies, Oak Ridge National Laboratory, 1950s). The numbers here are not from a specific study.

The new coat pattern: two kennels

Use: Glue under "The new coat pattern" · Copies: One per student

Claim: A chemical in Kennel A's water caused the new coat patterns.

Counterclaim: "New patterns are just copying errors. Copying errors happen in every dog."

Kennel	New coat patterns per 1,000 puppies
A (chemical in the water)	14
B (clean water)	3

Evidence: Kennel A had _____ new patterns per 1,000 puppies. Kennel B had _____.

Defense: Copying errors happen in _____ kennels, so they cannot explain the _____ extra patterns. Word bank: 11 · both · 3 · 14

Your turn: restate the claim, cite the data, and show why the other explanation does not fit.

Illustrative data: simplified numbers made up for class, not from a study.

UNIT 2.2 · LESSON 10 · CHECK MY CFAS

My target tracker

Use: Glue under "Check my CFAs" · Copies: One per student

Open Canvas. Write your CFA score for each target. Below a B (fewer than 4 of 5)? Do the review for that target. Then rate yourself 1 to 4 (4 = I can teach it).

Target	CFA score	Review done?	Self-rating (1 to 4)
LS3-2.1: I can name three ways new genetic variation arises.			
LS3-2.2: I can describe how making sex cells and fertilization mix alleles into new combinations.			
LS3-2.3: I can explain how copying errors and things like UV light change genes, and when that change is inherited.			
LS3-2.4: I can read data and make a claim, backed by that data, about what caused a genetic variation.			
LS3-2.5: I can defend my claim against another explanation using the evidence.			

Which target will I practice first, and how?

Hydrangea soil data

Use: Glue under "Enrichment: environment and traits" · Copies: One per student (enrichment)

A gardener grew copies of one hydrangea plant (the same DNA) in soils of different pH. In a garden of 100 plants from one parent, the soil varies:

Soil	Flower color	Plants
More acidic soil	Blue	42
Close to neutral soil	Purple	33
Less acidic (alkaline) soil	Pink	25

The plants have the same _____. Flower color varied because the _____ changed how the gene works. Word bank: soil · DNA

Is flower color here an inherited mutation? Explain with one piece of evidence.

_____

_____

_____

Science background: hydrangea sepal color depends on how much aluminum the plant can take up, which depends on soil pH (more acidic soil gives blue flowers; see Royal Horticultural Society plant advice on changing hydrangea flower colour). The plant counts are illustrative, made up for class.